

STACK

Q4)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 44
Problem Description: Dr. Malar was booking a tour package of IRCTC from Chennai to Delhi for his family. Two of the relatives was interested in joining to this tour. these two persons are studying engineering in computing technology. only one tickets are remaining in the IRCTC portal. So, Dr. Malar decided to book one ticket for out of those persons also along with his family members. she wants to identify the one person out of these persons. he decided to conduct a technical task to identify the right person to travel. the task was that, implement two stack operations in an array Can you help them to complete the task? Constraints $0 < n < 5$ only five elements has to be practiced for this operation first element pushed into stack1 second element pushed into stack2, likewise elements pushed into alternative stacks vice versa. Function Description - Create a data structure <i>twoStacks</i> that represents two stacks. - Implementation of <i>twoStacks</i> should use only one array, i.e., both stacks should use the same array for storing elements. - Following functions must be supported by <i>twoStacks</i>. push1(int x) -> pushes x to first stack push2(int x) -> pushes x to second stack pop1() -> pops an element from first stack and return the popped element					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int a[5];
5      int top1 = -1;
6      int top2 = 5;
7      int i, x;
8
9      for (i = 0; i < 5; i++) {
10         scanf("%d", &x);
11
12         if (i % 2 == 0) {
13             a[++top1] = x;
14         } else {
15             a[--top2] = x;
16         }
17     }
18
19     printf("Popped element from stack1 is:%d\n", a[top1--]);
20     printf("Popped element from stack2 is:%d", a[top2++]);
21
22     return 0;
23 }
```

OUTPUT:

TEST CASE 1:

```
1
2
3
4
5
Popped element from stack1 is:5
Popped element from stack2 is:4
```

TEST CASE 2:

```
5
4
5
3
4
Popped element from stack1 is:4
Popped element from stack2 is:3
```